

Commerce XI Maths App

Theory (Pages 40–42)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 22

Question 1 - Theory P40

If x is a cube root of unity, then

Options

A. $x^3 = 1$

B. $x^2 = 1$

C. $x^3 = 0$

D. $x = -1$ only

Correct answer: $x^3 = 1$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $x^3 = 1$.

Question 2 - Theory P40

Factor $x^3 - 1$

Options

A. $(x - 1)(x^2 + x + 1)$

B. $(x + 1)(x^2 - x + 1)$

C. $(x - 1)^3$

D. $(x - 1)(x^2 - x - 1)$

Correct answer: $(x - 1)(x^2 + x + 1)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(x - 1)(x^2 + x + 1)$.

Question 3 - Theory P40

The three cube roots of unity are

Options

A. $1, (-1 + \sqrt{3})i/2, (-1 - \sqrt{3})i/2$

B. $1, i, -i$

C. $1, -1, i$

D. $1, (1 + \sqrt{3})i/2, (1 - \sqrt{3})i/2$

Correct answer: $1, (-1 + \sqrt{3})i/2, (-1 - \sqrt{3})i/2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $1, (-1 + \sqrt{3})i/2, (-1 - \sqrt{3})i/2$.

Question 4 - Theory P40-41

If $w = (-1 + \sqrt{3}i)/2$, then w^2 equals

Options

- A. $(-1 - \sqrt{3}i)/2$
- B. $(-1 + \sqrt{3}i)/2$
- C. $(1 - \sqrt{3}i)/2$
- D. 1

Correct answer: $(-1 - \sqrt{3}i)/2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(-1 - \sqrt{3}i)/2$.

Question 5 - Theory P41

For a complex cube root of unity w , w^3 equals

Options

- A. 1
- B. 0
- C. -1
- D. w

Correct answer: 1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 1.

Question 6 - Theory P41

For $w \neq 1$, which relation holds?

Options

- A. $w^2 + w + 1 = 0$
- B. $w^2 - w + 1 = 0$
- C. $w^2 + w - 1 = 0$
- D. $w^2 = 1$

Correct answer: $w^2 + w + 1 = 0$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $w^2 + w + 1 = 0$.

Question 7 - Theory P41

From $w^2 + w + 1 = 0$, $w + w^2$ equals

Options

- A. -1
- B. 1
- C. 0
- D. w

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1 .

Question 8 - Theory P41

For a complex cube root of unity w , $(1)/(w)$ equals

Options

- A. w^2
- B. w
- C. 1
- D. $-w$

Correct answer: w^2

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w^2 .

Question 9 - Theory P41

For a complex cube root of unity w , $(1)/(w^2)$ equals

Options

- A. w
- B. w^2
- C. 1
- D. $-w^2$

Correct answer: w

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w .

Question 10 - Theory P41

For $n \in \mathbb{N}$, w^{3n} equals

Options

- A. 1
- B. w
- C. w^2
- D. 0

Correct answer: 1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 1.

Question 11 - Theory P41

For $n \in \mathbb{N}$, w^{3n+1} equals

Options

- A. w
- B. 1
- C. w^2
- D. $-w$

Correct answer: w

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w .

Question 12 - Theory P41

For $n \in \mathbb{N}$, w^{3n+2} equals

Options

- A. w^2
- B. 1
- C. w
- D. $-w^2$

Correct answer: w^2

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w^2 .

Question 13 - Theory P41

The conjugate of w is

Options

- A. w^2
- B. w
- C. 1
- D. $-w$

Correct answer: w^2

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w^2 .

Question 14 - Theory P41

The conjugate of w^2 is

Options

- A. w
- B. w^2
- C. 1
- D. $-w^2$

Correct answer: w

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives w .

Question 15 - Solved Example P41

Find $(1)/(w) + (1)/(w^2)$

Options

- A. -1
- B. 1
- C. 0
- D. -2

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1 .

Question 16 - Solved Example P41

Find $(1 + w^2)^3$

Options

- A. -1
- B. 1
- C. 0
- D. -8

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1.

Question 17 - Solved Example P41

Find $(1 - w + w^2)^3$

Options

- A. -8
- B. 8
- C. -1
- D. 0

Correct answer: -8

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -8.

Question 18 - Solved Example P41

Find $(1 - w)(1 - w^2)(1 - w^4)(1 - w^5)$

Options

- A. 9
- B. 3
- C. -9
- D. 0

Correct answer: 9

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 9.

Question 19 - Solved Example P41-42

If n is a multiple of 3, then $1 + w^n + w^{2n}$ equals

Options

- A. 3
- B. 0
- C. 1
- D. -1

Correct answer: 3

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 3.

Question 20 - Solved Example P41-42

If n is not a multiple of 3, then $1 + w^n + w^{2n}$ equals

Options

- A. 0
- B. 3
- C. 1
- D. -1

Correct answer: 0

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 0.

Question 21 - Theory P40

Among the cube roots of unity, the two non-real roots are

Options

- A. complex conjugates of each other
- B. equal real numbers
- C. additive inverses only
- D. both equal to i

Correct answer: complex conjugates of each other

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives complex conjugates of each other.

Question 22 - Theory P41

Which notation represents the cube roots of unity?

Options

A. $1, w, w^2$

B. $0, w, w^2$

C. $1, w, w^3$

D. $-1, w, w^2$

Correct answer: $1, w, w^2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $1, w, w^2$.